Advanced Machine Learning

DATA 642

Lecture 11 — GMMs for Anomaly Detection

*Figures and notes are from the book Mathematics for Machine Learning by A. Aldo Faisal, Cheng Soon Ong, and Marc Peter Deisenroth, Machine Learning A Bayesian and Optimization Perspective, Sergios Theodoridis, Elsevier, and Hands-On Machine Learning with Scikit-Learn Keras and TensorFlow, Aurelien Geron, O'Reilly

Anomaly Detection

- Anomaly detection, also known as outlier detection, is the task of identifying instances that deviate significantly from the norm. These instances, known as anomalies or outliers, exhibit behavior that is markedly different from the majority of the data, which are referred to as inliers.

- Anomaly detection finds applications across various domains, playing a crucial role in ensuring the integrity and reliability of systems and processes. Some common applications include:

    - **Fraud Detection:** Anomaly detection is widely used in fraud detection systems to identify suspicious transactions or activities that deviate from typical patterns.

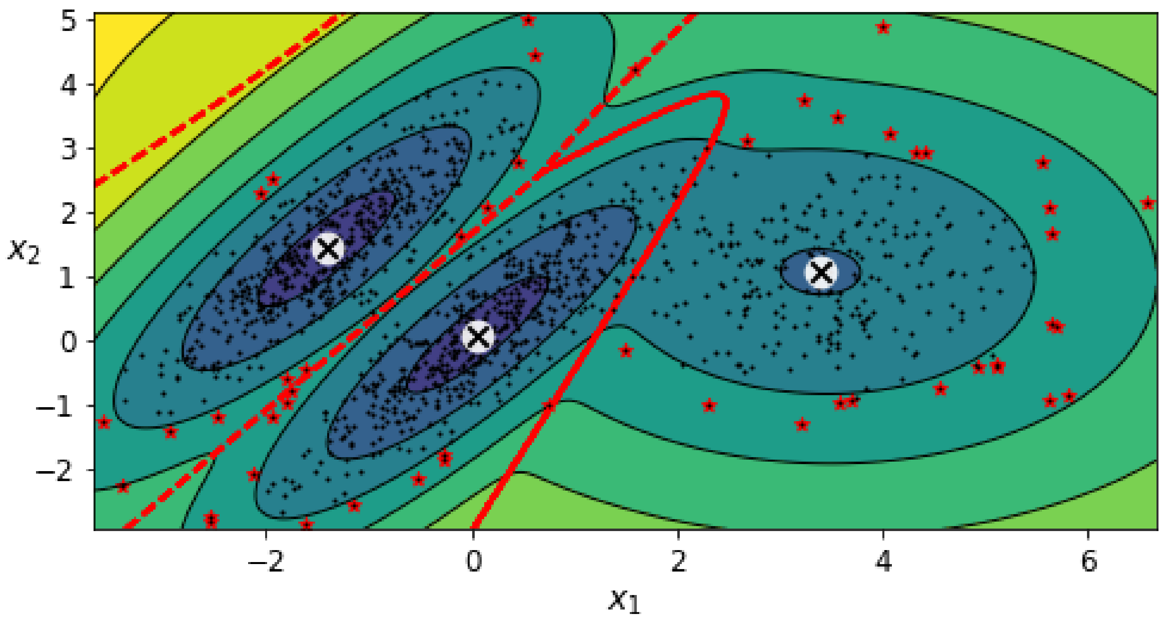
    - **Manufacturing Quality Control:** In manufacturing environments, anomaly detection is employed to detect defective products or equipment failures.

    - **Data Preprocessing:** Anomaly detection is often used as a preprocessing step in data analysis pipelines to identify and remove outliers from datasets. By filtering out anomalous data points before training machine learning models, the resulting models can be more robust and accurate, as they are less influenced by noisy or erroneous data.

Gaussian Mixture Models (GMMs) for Anomaly Detection

- **Modeling Normal Behavior:**
    - GMMs can effectively model the normal behavior of data using multivariate Gaussian distributions. Any deviations from this behavior can be flagged as anomalies.
- **Flexibility in Modeling Complex Distributions:**
    - GMMs offer flexibility in capturing complex data distributions, making them suitable for detecting anomalies in diverse datasets. This includes anomalies in network traffic, financial transactions, or sensor data.

- **Density-based Anomaly Detection:**
  - GMMs can be leveraged for density-based anomaly detection. Instances located in low-density regions of the data distribution can be considered anomalies.
  - To define the density threshold for anomaly detection, one can utilize domain knowledge or set it based on a specific ratio of anomalies. For example, in manufacturing, if the known ratio of defective products is 4%, the density threshold can be set accordingly to capture 4% of instances as anomalies. Anomalies here are denoted with stars.



Limitations:

1. **Assumption of Gaussian Distribution:**
    - GMMs assume that the data is generated from a mixture of Gaussian distributions. If the data does not conform to this assumption, GMMs may struggle to accurately model the underlying distribution, leading to suboptimal anomaly detection performance.
2. **Sensitive to Initialization:**
    - GMMs are sensitive to initialization, and the choice of initial parameters can significantly impact the results. Poor initialization may lead to convergence to suboptimal solutions or convergence to local optima.

3. **Difficulty in Capturing Non-linear Relationships:**
    - GMMs inherently model linear relationships between features. Therefore, they may struggle to capture complex non-linear relationships present in the data. This limitation can result in poor performance when detecting anomalies in datasets with non-linear structures.
4. **Difficulty with High-dimensional Data:**
    - GMMs may face challenges when dealing with high-dimensional data. Estimating parameters in high-dimensional spaces requires a large amount of data, and GMMs may suffer from the curse of dimensionality, leading to overfitting or computational inefficiency.
5. **Biased model**
    - Gaussian mixture models try to fit all the data, including the outliers, so if you have too many of them, this will bias the model's view of "normality": some outliers may wrongly be considered as normal.

Tips and Considerations:

1. **Feature Selection and Dimensionality Reduction:**
    - Before applying GMMs for anomaly detection, carefully select relevant features and consider using dimensionality reduction techniques such as Principal Component Analysis (PCA) to reduce the dimensionality of the data. This can help mitigate the curse of dimensionality and improve model performance.
2. **Model Evaluation and Validation:**
    - Evaluate the performance of the GMM-based anomaly detection algorithm using appropriate metrics such as precision, recall, and F1-score. Additionally, validate the model using cross-validation or holdout sets to ensure its generalization ability.
3. **Tune Model Hyperparameters:**
    - Experiment with different hyperparameters such as the number of mixture components, covariance type, and regularization parameters.